UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 7E – Predetermined Motion Time Systems II

EIN3241: Human Factors and Ergonomics
Spring 2026

April 8, 2026

Dr. Jad A. Atweh

Admin Issues

- Final Exam – Extra Credit Assignment
  - 5 points
  - Due Monday April 27 11:59 PM

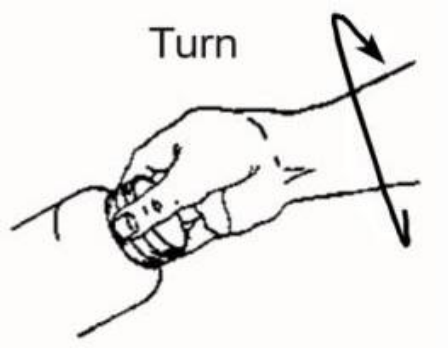
- Quiz 6: Monday or Wednesday?

MTM-1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
     - Reach
     - Move
     - Turn
     - Apply pressure
     - Grasp
     - Position
     - Disengage
     - Release
     - Eye travel time and focus
     - Body, leg, and foot motions
  5. Remove the non-limiting values
  6. Calculate standard time

Turn

- Turn is a movement that rotates the hand and wrist. It depends on
  - The number of degrees turned
  - The weight of the object or the resistance against which the turn is made
    - Small (S)
    - Medium (M)
    - Large (L)



T30S

T45M

Table III—Turn and Apply Pressure—T and AP											
	Time TMU for Degrees Turned										
Weight	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Small—0 to 2 lb	2.8	3.5	4.1	4.8	5.4	6.1	6.8	7.4	8.1	8.7	9.4
Medium—2.1 to 10 lb	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large—10.1 to 35 lb	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

Apply Pressure Case A—10.6 TMU. Apply Pressure Case B—16.2 TMU

Apply pressure

- Apply pressure is the application of force without resultant movement
  - Case A: application of pressure
  - Case B: application of pressure with higher precision
    - Use when you need to hold something carefully in place
    - Use when there is a position before applying pressure

Table III—Turn and Apply Pressure—T and AP											
	Time TMU for Degrees Turned										
Weight	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Small—0 to 2 lb	2.8	3.5	4.1	4.8	5.4	6.1	6.8	7.4	8.1	8.7	9.4
Medium—2.1 to 10 lb	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large—10.1 to 35 lb	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

Apply Pressure Case A—10.6 TMU. Apply Pressure Case B—16.2 TMU

AP
A

APB

Grasp

- Gain control of an object; often followed by Move.
  1. Pick up grasp: grasp an individual object
     A. By itself (most common)
     B. Small object lying against the surface (like a small paper)
     C. Cylindrical objects (like pens or markers next to each other)
  2. Regrasp: change grasp without relinquishing control; object stays in your hand
  3. Transfer: control transferred from one hand to the other
  4. Select: objects jumbled with each other such that search and select often occur
     A. Large
     B. Medium
     C. Small
  5. Contact: contact or sliding grasp (like pressing a button); your fingers to do not really hold on to anything

Grasp

G1C2

Table IV—Grasp—G		
Case	**Time (TMU)**	**Description**
1A	2.0	**Pick Up Grasp**—Small, medium or large object by itself, easily grasped.
1B	3.5	Very small object or object lying close against a flat surface.
1C1	7.3	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter larger than $^1/_2''$.
1C2	8.7	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter $^1/_4''$ to $^1/_2''$.
1C3	10.8	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter less than $^1/_4''$.
2	5.6	**Regrasp.**
3	5.6	**Transfer Grasp.**
4A	7.3	Object jumbled with other objects so search and select occur. Larger than $1'' \times 1'' \times 1''$.
4B	9.1	Object jumbled with other objects so search and select occur. $^1/_4'' \times {}^1/_4'' \times {}^1/_8''$ to $1'' \times 1'' \times 1''$.
4C	12.9	Object jumbled with other objects so search and select occur. Smaller than $^1/_4'' \times {}^1/_4'' \times {}^1/_8''$.
5	0	Contact, sliding or hook grasp.

Position

- Align, orient, and engage one object with another object. Depends on:
  - Class of fit (loose, close, exact):
    - Loose: like a cup to its saucer
    - Close: like a marker cap to a marker
    - Exact: like a key to its keyhole
  - Symmetry (symmetrical, semi-symmetrical, non- symmetrical)
    - Symmetrical: infinity of ways to insert the object (like a pen inside a cup)
    - Semi-symmetrical: finite ways to insert the object (like a paper into a rectangular cardboard box)
    - Non-symmetrical: one and only one way to insert the object (like a key inside a keyhole)
  - Ease of handling: how easy the object is to hold (slippery, dangerous, sharp…)

P1SE

Table V—Position*—P

Class of Fit		Symmetry	Easy to Handle	Difficult to Handle
1—Loose	No pressure required.	S	5.6	11.2
		SS	9.1	14.7
		NS	10.4	16.0
2—Close	Light pressure required.	S	16.2	21.8
		SS	19.7	25.3
		NS	21.0	26.6
3—Exact	Heavy pressure required.	S	43.0	48.6
		SS	46.5	52.1
		NS	47.8	53.4

*Distance moved to engage—1″ or less.

Disengage

- Break contact between one object and another
  - Case 1: Loose
  - Case 2: Close
  - Case 3: Tight

Table 13.1 (continued)

Table VII—Disengage—D		
Class of Fit	**Easy to Handle**	**Difficult to Handle**
1—Loose—Very slight effort, blends with subsequent move.	4.0	5.7
2—Close—Normal effort, slight recoil.	7.5	11.8
3—Tight—Considerable effort, hand recoils markedly.	22.9	34.7

D1E

Release

- The relinquishing of control of an object by the hand or finger
  - Case 1: normal release by opening of fingers
  - Case 2: contact release (always for a contact grasp)

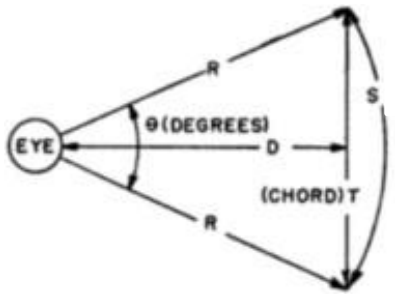
RL1

Table VI—Release—RL

Case	Time (TMU)	Description
1	2.0	Normal release performed by opening fingers as independent motion.
2	0	Contact release.

Eye travel time and focus

- Eye travel time = time to move the eyes = 15.2 (T/D) TMU
  - Where T is the distance traveled by the eye and D is the perpendicular distance from the eye to the line of travel



  - Maximum of 20 TMU
  - Only use when explicitly stated
  - Always place in right hand column
  - Symbol: ET(T/D)

Eye travel time and focus

- Eye focus time: eye activity that takes place in order to determine easily noticeable characteristics of an object within the normal field of vision
  - Constant: 7.3 TMU
  - Often comes up in inspections or reading a word
  - Always place in right hand column

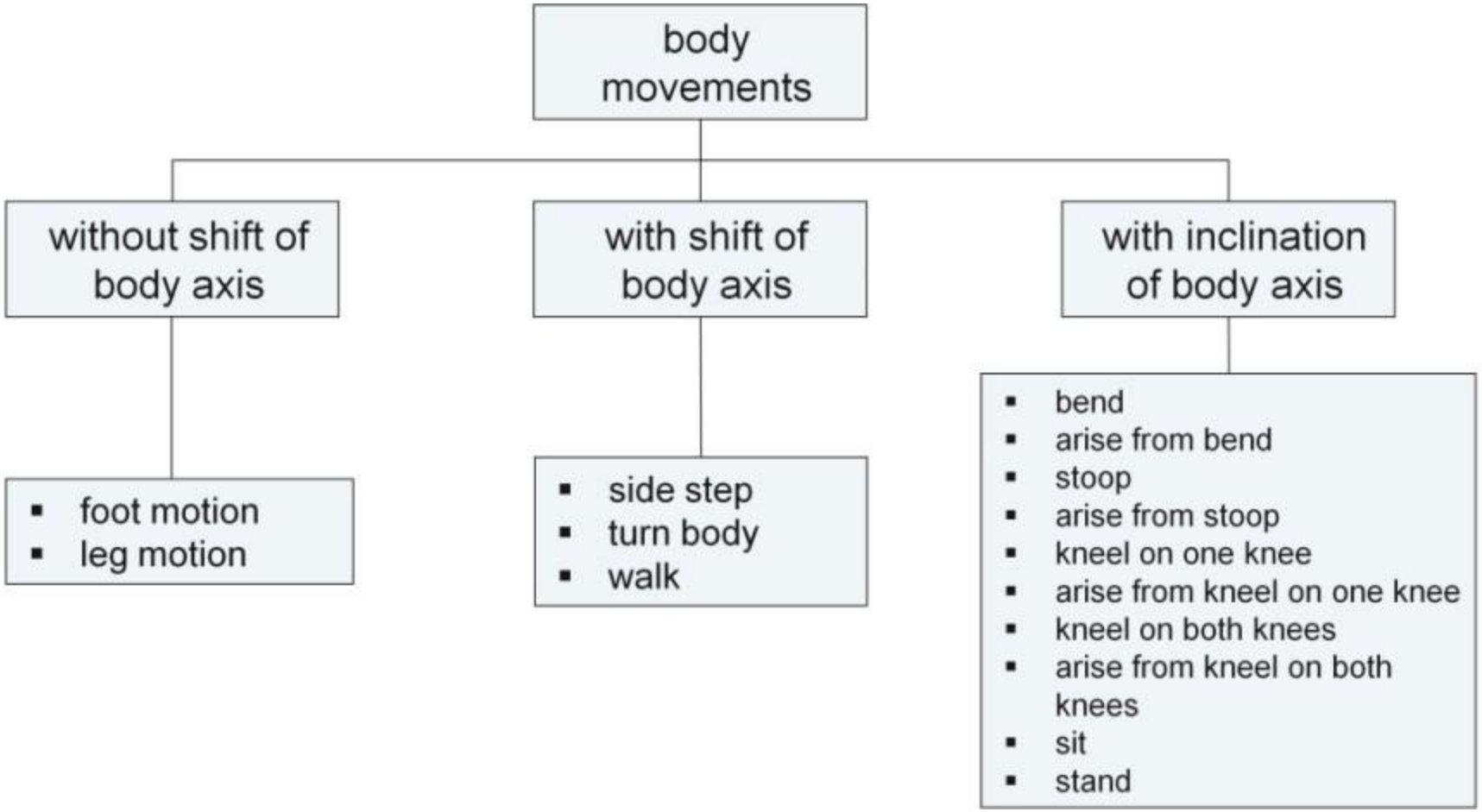
Table VIII—Eye Travel Time and Eye Focus—ET and EF

Eye Travel Time $= 15.2 \times \frac{T}{D}$ TMU, with a maximum value of 20 TMU.

where T = the distance between points from and to which the eye travels.
D = the perpendicular distance from the eye to the line of travel T.

Eye Focus Time = 7.3 TMU.

Body, leg, and foot motions



Body, leg, and foot motions

- Foot motion: like pressing the brakes
- Leg or foreleg motion: like swinging your legs while on a swing
- Sidestep:
  - Case 1: move one leg to the side
  - Case 2: move one leg to the side, then bring the other leg to it
- Bend: bow down without bending the knees
- Stoop: bend your knees so you can touch the floor

Body, leg, and foot motions

Table IX—Body, Leg, and Foot Motions			
Description	**Symbol**	**Distance**	**Time TMU**
Foot Motion—Hinged at ankle.	FM	Up to 4″	8.5
With heavy pressure.	FMP		19.1
Leg or Foreleg Motion	LM—	Up to 6″	7.1
		Each add'l. inch	1.2
Sidestep—Case 1—Complete when leading leg contacts floor.	SS-C1	Less than 12″	Use REACH or MOVE time
		12″	17.0
		Each add'l. inch	0.6
Case 2—Lagging leg must contact floor before next motion can be made.	SS-C2	12″	34.1
		Each add'l. inch	1.1
Bend, Stoop, or Kneel on One Knee.	B, S, KOK		29.0
Arise.	AB, AS, AKOK		31.9
Kneel on Floor—Both Knees.	KBK		69.4
Arise.	AKBK		76.7
Sit.	SIT		34.7
Stand from Sitting Position.	STD		43.4
Turn Body 45 to 90 degrees—			
Case 1—Complete when leading leg contacts floor.	TBC1		18.6
Case 2—Lagging leg must contact floor before next motion can be made.	TBC2		37.2
Walk.	W-FT.	Per foot	5.3
Walk.	W-P	Per pace	15.0

One leg changes position

Two legs change position

Arise from bend, stoop, kneel

Example

- Put bolt in hole

Left-hand description	F	LH motion	TMU	RH motion	F	Right-hand description
Reach for bolt 32 inches away that is jumbled up with others						
Grasp bolt whose largest dimension is 0.3 inches						
Move the bolt to the apparatus hole 20 inches away						
Position the bolt in the hole						
Release bolt						

Tables

Add 0.8 TMU per inch for each additional inch

Table 13.1 Summary of MTM-1 Data

Table I—Reach—R

Distance Moved (in)	Time (TMU) A	B	C or D	E	Hand in Motion A	B
1/2 or less	2.0	2.0	2.0	2.0	1.6	1.6
1	2.5	2.5	3.6	2.4	2.3	2.3
2	4.0	4.0	5.9	3.8	3.5	2.7
3	5.3	5.3	7.3	5.3	4.5	3.6
4	6.1	6.4	8.4	6.8	4.9	4.3
5	6.5	7.8	9.4	7.4	5.3	5.0
6	7.0	8.6	10.1	8.0	5.7	5.7
7	7.4	9.3	10.8	8.7	6.1	6.5
8	7.9	10.1	11.5	9.3	6.5	7.2
9	8.3	10.8	12.2	9.9	6.9	7.9
10	8.7	11.5	12.9	10.5	7.3	8.6
12	9.6	12.9	14.2	11.8	8.1	10.1
14	10.5	14.4	15.6	13.0	8.9	11.5
16	11.4	15.8	17.0	14.2	9.7	12.9
18	12.3	17.2	18.4	15.5	10.5	14.4
20	13.1	18.6	19.8	16.7	11.3	15.8
22	14.0	20.1	21.2	18.0	12.1	17.3
24	14.9	21.5	22.5	19.2	12.9	18.8
26	15.8	22.9	23.9	20.4	13.7	20.2
28	16.7	24.4	25.3	21.7	14.5	21.7
30	17.5	25.8	26.7	22.9	15.3	23.2

Case and Description

A Reach to object in fixed location, or to object in other hand or on which other hand rests.

B Reach to single object in location which may vary slightly from cycle to cycle.

C Reach to object jumbled with other objects in a group so that search and select occur.

D Reach to a very small object or where accurate grasp is required.

E Reach to indefinite location to get hand in position for body balance or next motion or out of way.

Table II—Move—M

Distance Moved (in)	Time (TMU) A	B	C	Hand in Motion B	Wt. Allowance Wt. (lb) up to	Factor	Constant (TMU)	Case and Description
1/2 or less	2.0	2.0	2.0	1.7	2.5	0	0	A Move object to other hand or against stop.
1	2.5	2.9	3.4	2.3				
2	3.6	4.6	5.2	2.9	7.5	1.06	2.2	
3	4.9	5.7	6.7	3.6				
4	6.1	6.9	8.0	4.3	12.5	1.11	3.9	
5	7.3	8.0	9.2	5.0				
6	8.1	8.9	10.3	5.7	17.5	1.17	5.6	
7	8.9	9.7	11.1	6.5				
8	9.7	10.6	11.8	7.2				B Move object to approximate or indefinite location.
9	10.5	11.5	12.7	7.9	22.5	1.22	7.4	
10	11.3	12.2	13.5	8.6				
12	12.9	13.4	15.2	10.0	27.5	1.28	9.1	
14	14.4	14.6	16.9	11.4				
16	16.0	15.8	18.7	12.8	32.5	1.33	10.8	
18	17.6	17.0	20.4	14.2				
20	19.2	18.2	22.1	15.6	37.5	1.39	12.5	
22	20.8	19.4	23.8	17.0				
24	22.4	20.6	25.5	18.4	42.5	1.44	14.3	C Move object to exact location.
26	24.0	21.8	27.3	19.8				
28	25.5	23.1	29.0	21.2	47.5	1.50	16.0	
30	27.1	24.3	30.7	22.7				

Table IV—Grasp—G

Case	Time (TMU)	Description
1A	2.0	**Pick Up Grasp**—Small, medium or large object by itself, easily grasped.
1B	3.5	Very small object or object lying close against a flat surface.
1C1	7.3	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter larger than 1/2″.
1C2	8.7	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter 1/4″ to 1/2″.
1C3	10.8	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter less than 1/4″.
2	5.6	**Regrasp.**
3	5.6	**Transfer Grasp.**
4A	7.3	Object jumbled with other objects so search and select occur. Larger than 1″ × 1″ × 1″.
4B	9.1	Object jumbled with other objects so search and select occur. 1/4″ × 1/4″ × 1/8″ to 1″ × 1″ × 1″.
4C	12.9	Object jumbled with other objects so search and select occur. Smaller than 1/4″ × 1/4″ × 1/8″.
5	0	Contact, sliding or hook grasp.

Table V—Position*—P

Class of Fit		Symmetry	Easy to Handle	Difficult to Handle
1—Loose	No pressure required.	S	5.6	11.2
		SS	9.1	14.7
		NS	10.4	16.0
2—Close	Light pressure required.	S	16.2	21.8
		SS	19.7	25.3
		NS	21.0	26.6
3—Exact	Heavy pressure required.	S	43.0	48.6
		SS	46.5	52.1
		NS	47.8	53.4

*Distance moved to engage—1″ or less.

Table VI—Release—RL

Case	Time (TMU)	Description
1	2.0	Normal release performed by opening fingers as independent motion.
2	0	Contact release.

Example – Answers

- Put bolt in hole

Left-hand description	F	LH motion	TMU	RH motion	F	Right-hand description
Reach for bolt 32 inches away that is jumbled up with others		R32C	28.3			
Grasp bolt whose largest dimension is 0.3 inches		G4B	9.1			
Move the holt to the apparatus hole 20 inches away		M20C	22.1			
Position the bolt in the hole		P2SE	16.2			
Release bolt		RL1	2			

MTM–1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
  5. Remove the non-limiting values
  6. Calculate standard time

Remove non-limiting values

- Analyze the fundamental motions
  1. Align the LH and RH motions in terms of time
     - Keep the order and alignment, regardless of the time
  2. If there are no simultaneous motions, move to step 6 (standard time)
  3. If there are simultaneous motions: check if they can be done at the same time
     - If yes, circle the longer motion (limiting) and consider that time only
     - If no, circle both motions and consider both times

Assumed to be easy unless novice specified

If there are two options (w/o), assume worse option

Table X—Simultaneous Motions

Reach			Move				Grasp				Position			Disengage			
A, E	B	C, D	A, Bm	B	C		G1A G2 G5	G1B G1C	G4		P1S	P1SS P2S	P1NS P2SS P2NS	D1E D1D	D2	Case	Motion
		*w|o	*w|o	*w|o	*w|o			*w|o	*w|o		**E|D	**E|D	**E|D		**E|D		
□	□	□□	□□	□□	□□	××	□	□□	□□		□□	□×	××	□	□□	A, E	Reach
□	□	□×	□□	□□	□×	×■	□	□×	×■		××	×■	■■	□	□×	B	Reach
		□×	×■	×■	■■		□	×■	■■		■■	■■	■■	×	■■	C, D	Reach
			□□	□□	□□		□	□□	□□		□□	□×	××	□	□□	A, Bm	Move
				□□	□□		□	□×	×■		××	×■	■■	□	□×	B	Move
					×■		□	×■	■■		■■	■■	■■	×	■■	C	Move
							□	□□	□□		□□	□■	■■	□	■■	G1A, G2, G5	Grasp
								■■	×■		■■	■■	■■	■	■■	G1B, G1C	Grasp
									■■		■■	■■	■■	■	■■	G4	Grasp
											×■	■■	■■	■	■■	P1S	Position
												■■	■■	■	■■	P1SS, P2S	Position
													■■	■	■■	P1NS, P2SS, P2NS	Position
														□	□□	D1E, D1D	Disengage
															□□	D2	Disengage

□ = EASY to perform simultaneously.
× = Can be performed simultaneously with PRACTICE.
■ = DIFFICULT to perform simultaneously even after long practice. Allow both times.

MOTIONS NOT INCLUDED IN ABOVE TABLE
TURN—Normally EASY with all motions except when TURN is controlled or with DISENGAGE.
APPLY PRESSURE—May be EASY, PRACTICE, or DIFFICULT. Each case must be analyzed.
POSITION—Class 3—Always DIFFICULT.
DISENGAGE—Class 3—Normally DIFFICULT.
RELEASE—Always EASY.
DISENGAGE—Any class may be DIFFICULT if care must be exercised to avoid injury or damage to object.
*W = Within the area of normal vision.
O = Outside the area of normal vision.
**E = EASY to handle.
D = DIFFICULT to handle.

Source: MTM Association for Standards and Research, Fair Lawn, New Jersey.

- For body motions, always put in right hand and analyze on a case-by- case basis
- Same for eye motions; need to consider where person is looking

Remove non-limiting values

- Consider the following example:

Left-hand description	F	LH motion	TMU	TMU	RH motion	F	Right-hand description
Reach hand to bin		R10C	12.9	1.0	R1L		Release assembly
Grasp washer		G4A	8.7	8.4	R4C		Reach to bin
Move to fixture		M8C	11.8	8.4	G4B		Grasp pin
Position washer		P1SE	5.6	7.3	M4C		Move to fixture

MTM–1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
  5. Remove the non-limiting values
  6. Calculate standard time

Calculate standard time

- Sum all the individual (non-simultaneous) motions and all the circled simultaneous motions
- Final result is in TMU
- Can convert to
  - Seconds: x0.036
  - Minutes: x0.0006
  - Hours: x0.00001
- Apply the appropriate allowance to obtain the standard time

PTS

- MTM-1: first PTS system
  - Very detailed and specific
  - Includes all kinds of motions
  - Can help design work that is not set up
- Problems with MTM-1
  - Very time consuming and tedious
  - Very generic

UF
Herbert Wertheim
College of Engineering
UNIVERSITY of FLORIDA
LEADING THE CHARGE, CHARGING AHEAD